

Secure Static Website Hosting using Private S3 and CloudFront

Introduction

A Secure Static Website architecture allows hosting static web content while preventing direct public access to Amazon S3. The website files are stored in a private S3 bucket, and users access the content through Amazon CloudFront. This approach improves security, performance, and scalability.

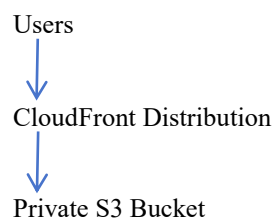
Objective

The objective of this project is to host a static website securely using a private Amazon S3 bucket and Amazon CloudFront with Origin Access Control (OAC), ensuring that website content can only be accessed through CloudFront.

AWS Services Used

- Amazon S3
- Amazon CloudFront
- Origin Access Control (OAC)
- IAM Policies
- AWS Global Edge Locations

Architecture



Working Principle

1. Website files are stored in a private S3 bucket.
2. CloudFront is configured as the content delivery network.
3. Origin Access Control (OAC) securely connects CloudFront to S3.
4. Bucket policies allow access only from CloudFront.
5. Users access website content through the CloudFront URL.
6. Direct access to S3 objects remains blocked.

What is Origin Access Control (OAC)?

Origin Access Control (OAC) is a CloudFront feature that securely signs requests sent to Amazon S3. It ensures that only CloudFront can access objects stored in a private S3 bucket.

Benefits

- Enhanced Security
- Private S3 Storage
- Global Content Delivery
- Reduced Latency
- Improved Performance
- Cost Optimization through Caching

Security Implementation

The S3 bucket remains private with Block Public Access enabled. CloudFront is granted permission through OAC and a bucket policy that restricts access exclusively to the CloudFront distribution.

Verification

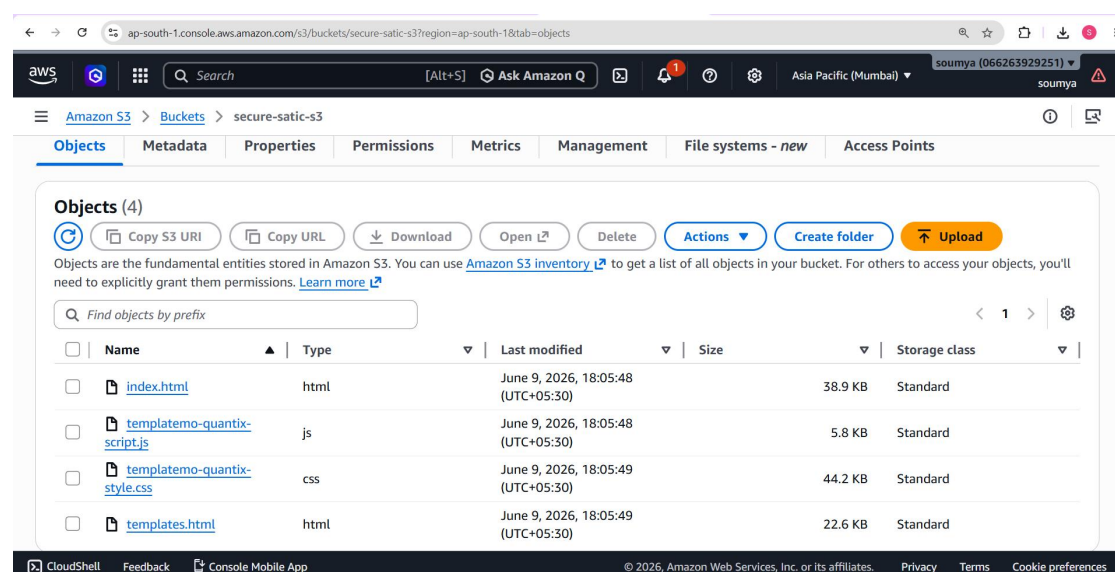
Successful implementation is verified when:

- Website loads through CloudFront.
- Direct S3 object URLs return Access Denied.
- Bucket remains private.

Conclusion

This project demonstrates a secure and scalable static website hosting architecture using Amazon S3 and CloudFront. By implementing Origin Access Control, website content is protected while still being globally accessible through CloudFront.

Screenshots:



ap-south-1.console.aws.amazon.com/s3/buckets/secure-satic-s3?region=ap-south-1&tab=permissions

Amazon S3 > Buckets > secure-satic-s3

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access
On

► Individual Block Public Access settings for this bucket

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about [using Amazon S3 Block Public Access](#)

us-east-1.console.aws.amazon.com/cloudfront/v4/home?region=ap-south-1#/distributions/E3M73NOBAEIH7A

CloudFront > Distributions > E3M73NOBAEIH7A

Successfully updated distribution settings.

secure-static-cf

View metrics

Details

Distribution domain name d1z1xjashv892.cloudfront.net	Billing	ARN arn:aws:cloudfront::066263929251:distribution/E3M73NOBAEIH7A	Last modified June 9, 2026 at 12:39:46 PM UTC
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Settings

Name secure-static-cf	Alternate domain names Add domain	Standard logging Off
Description -		Cookie logging Off
Price class Use all edge locations (best performance)		Default root object index.html
Supported HTTP versions HTTP/2, HTTP/1.1, HTTP/1.0		Cache tag header -

d1z1xjashv892.cloudfront.net

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Traffic - Last 30 Days

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OVERVIEW: Implementing Secure Static Website Hosting with Amazon S3, CloudFront & OAC

